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PERSUASION: EMPIRICAL EVIDENCE

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ABSTRACT

We provide a selective survey of empirical evidence on the effects as well as the drivers of persuasive communication. We consider persuasion directed at consumers, voters, donors, and investors. We organize our review around four questions. First, to what extent does persuasion affect the behavior of each of these groups? Second, what models best capture the response to persuasive communication? In particular, we distinguish information-based models from preference-based models. Third, what are persuaders' incentives and what limits their ability to distort communications? Finally, what evidence exists on the equilibrium outcomes of persuasion in economics and politics?

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Abstract

We provide a selective survey of empirical evidence on the effects as well as the drivers of persuasive communication. We consider persuasion directed at consumers, voters, donors, and investors. We organize our review around four questions. First, to what extent does persuasion affect the behavior of each of these groups? Second, what models best capture the response to persuasive communication? In particular, we distinguish information-based models from preference-based models. Third, what are persuaders' incentives and what limits their ability to distort communications? Finally, what evidence exists on the equilibrium outcomes of persuasion in economics and politics?

Keywords: Persuasion, Communication, Beliefs

1 Introduction

The efficiency of market economies and democratic political systems depends on the accuracy of individuals' beliefs. Consumers need to know the set of goods available and be able to evaluate their characteristics. Voters must understand the implications of alternative policies and be able to judge the likely behavior of candidates once in office. Investors should be able to assess the expected profitability of firms.

Some beliefs are shaped by direct observation. But a large share of the information on which economic and political decisions are based is provided by agents who themselves have an interest in the outcome. Information about products is delivered through advertising by the sellers, political information comes from candidates interested in winning election, and financial data is released strategically to shape the perceptions of investors. Third parties that might be more objective—certifiers, media firms, financial analysts—have complex incentives that may diverge from the interests of recipients.

Scholars have reached sharply different conclusions about the implications of communication by motivated agents. Much of the early writing—shaped by the growth of mass advertising in the late nineteenth century and the powerful propaganda campaigns of the early twentieth—is pessimistic. Lippmann (1922), for example, sees citizens as largely at the mercy of powerful persuaders. Democracy asks them to express opinions on topics about which they can have no direct experience, and the fact that their views are shaped by those in power undermines democracy's basic premise. Robinson (1933), Kaldor (1950), Galbraith (1967), and others view advertising as anti-competitive and detrimental to welfare.

The alternative view is that persuasive communication is a more or less efficient provision of information. That those providing the information have ulterior motives may be no more troubling than the fact that Adam Smith's butchers and bakers act out of their own self interest. Bernays (1928) defends the value of propaganda on the grounds that even distorted communications are often informative. The influential work of Stigler (1961) and Telser (1964) shows that in a world of incomplete information advertising can enhance market

efficiency. Downs (1957) sees political campaigns and media coverage playing a similar role, delivering information at low cost to voters who would have insufficient incentive to seek it out themselves.

In this review, we provide a selective survey of empirical evidence on the effects as well as the drivers of persuasive communication. We define a persuasive communication to be a message provided by one agent (a sender) with at least a potential interest in changing the behavior of another agent (a receiver). We exclude situations where an agent attempts to manipulate another through means other than messages—providing monetary incentives, for example, or outright coercion. We also depart from some existing literatures in using the term persuasion to refer to both informative and non-informative dimensions of messages.

Some of the evidence we review comes from economists, including a recent surge of work applying modern empirical tools. Other evidence comes from marketing, political science, psychology, communications, accounting, and related fields. Our main focus is on studies that credibly identify effects on field outcomes. We do not attempt to review the laboratory evidence on persuasion, nor do we discuss literatures where the outcome measures are primarily self-reported attitudes. Even within these parameters, we inevitably omit many important studies, and give only cursory treatment to others.

We consider four groups that are targets of persuasion: consumers, voters, donors, and investors. We organize our review around four questions. First, to what extent does persuasion affect the behavior of each of these groups? Second, what models best capture the response to persuasive communication? In particular, we distinguish belief-based models from preference-based models. Third, what are persuaders' incentives and what limits their ability to distort communications? Finally, what evidence exists on the equilibrium outcomes of persuasion in economics and politics?

2 Is Persuasion Effective?

The simplest question addressed by studies in this literature is whether persuasive messages have a measurable effect on receiver behavior. Credibly measuring the direct effect of persuasion is both important in its own right and an input to studying finer questions about the way persuasion effects vary with sender incentives, receiver information, and so forth.

Wherever possible, we report results in terms of the “persuasion rate” (DellaVigna and Kaplan 2007), which estimates the percentage of receivers that change the behavior among those that receive a message and are not already persuaded. In a setting with a binary behavioral outcome, a treatment group T , and a control group C , the persuasion rate f (in percent terms) is

$$f = 100 * \frac{y_T - y_C}{e_T - e_C} \frac{1}{1 - y_0}, \quad (1)$$

where e_i is the share of group i receiving the message, y_i is the share of group i adopting the behavior of interest, and y_0 is the share that would adopt if there were no message. Where y_0 is not observed, we approximate it by y_C .¹

The persuasion rate captures the effect of the persuasion treatment on the relevant behavior ($y_T - y_C$), adjusting for exposure to the message ($e_T - e_C$) and for the size of the population left to be convinced ($1 - y_0$). For example, suppose that get-out-the-vote mailers sent to the treatment group increase turnout by 1 percentage point relative to the control group. The size of this effect may suggest that mailers are not very persuasive. However, the implied persuasion rate can in reality be high if, say, only 10 percent of voters in the treatment group received a get-out-the vote mailer ($e_T - e_C$ is small) and the targeted population already had high turnout rates of 80 percent ($1 - y_0$ is small). In this case, the implied persuasion rate is $f = 100 * (.01) / (.1 * .2) = 50$ percent. If the persuadable population is small, a small change in behavior can imply a high impact of persuasion.

¹Assuming random exposure and a constant persuasion rate f , the share in group i who adopt the behavior is $y_i = y_0 + e_i f (1 - y_0)$. Rearranging this expression gives equation 1. Solving the system for y_0 , one obtains $y_0 = y_C - e_C (y_T - y_C) / (e_T - e_C)$. The approximation $y_0 = y_C$ is valid as long as the exposure rate in the control group is small ($e_C \approx 0$) or the effect of the treatment is small ($y_T - y_C \approx 0$).

Table 1 summarizes persuasion rates for the studies discussed below.

2.1 Persuading Consumers

The first form of persuasion we consider is communication directed at consumers. Most studies in this domain focus on communication from firms in the form of advertising. The behavioral outcome is typically purchases or sales volume.

The earliest studies of advertising effects exploit messages that can be linked directly to receiver responses. Shryer (1912), for example, places classified ads for a mail-order business that include a code customers use when they reply. He compiles data on the response rates of each ad he placed, and analyzes their relative effectiveness and the extent of diminishing returns. This method is not ideal for measuring persuasion rates, since ads are not randomized and the counter-factual responses in the absence of an ad are not observed. The same methodology can be used to study online search advertising where the response rate to individual ads is measurable directly.

Field experiments have long been used to study advertising effects, with many studies carried out by advertisers themselves. Hovde (1936), for example, discusses “split-run” experiments conducted in cooperation with the Chicago Tribune in which groups of subscribers are assigned to receive different advertising inserts in their Sunday papers. Similar split-run tests are discussed by West (1939), Zubin and Peatmen (1943), and Waldman (1956). Ackoff and Emshoff (1975) report on one of the earliest television advertising experiments—a series of tests carried out by Anheuser-Busch between 1962 and 1964 in which different markets are assigned different advertising levels. A large number of other market-level experiments are reported by Aaker and Carman (1982), Eastlack and Rao (1989) and Hu, Lodish, and Krieger (2007). Results using individual-level assignment of advertising exposure are reported by McGuire (1977), Lodish et al. (1995), and Hu, Lodish, and Krieger (2007).

Overall, the results of these marketing studies provide little support for the view that there is a consistent effect of advertising spending on sales. The Anheuser Busch studies find

that both increasing *and decreasing* advertising increased sales; these results led the firm to significantly cut advertising in most of its markets. Aaker and Carman (1982) and Lodish et al. (1995) review large numbers of experiments and conclude that a large share of ads have no detectable effect. Hu, Lodish, and Krieger (2007) is the only review of experimental evidence that detects positive effects of advertising spending on average. The results of all these studies must be interpreted with caution, because the assignment to treatment and control conditions is often not randomized explicitly, and because the reported results often lack sufficient detail to assess the quality of the statistical analysis. Nevertheless, the negative results are striking given that the most obvious biases would tend to overstate the effect of advertising.

A series of recent papers by economists use field experiments to estimate the effect of persuasive communication on sales. The results, like those in the marketing literature, are mixed. Simester et al. (2007) report a field experiment in which they randomly vary the number of catalogs individuals receive by mail from a women’s clothing retailer. They find that increasing the number of catalogs in an 8-month period from 12 to 17 increases the number of purchases during the test period by 5 percent for customers who had purchased frequently in the past, and by 14 percent for those who had purchased relatively infrequently. The effect on the extensive margin (the share of customers who purchase at least one item) implies a higher persuasion rate percent for the frequent buyers ($f = 7.0$ percent) than for the less frequent buyers ($f = 4.3$ percent) (Table 1).² Lewis and Reiley (2009) report the results of an online advertising experiment involving more than a million and a half users of Yahoo whose purchases were also tracked at a large retailer. These subjects were randomly assigned to either a treatment group that saw ads for the retailers or a control group that did not. Because the ads were only shown on certain websites, only 64 percent of the treatment group was actually shown ads. The purchases of the treatment group were 3

²In this case, we do not observe how many of the customers in the treatment group received and read the catalogs, that is, we do not observe e_T . In this and in similar subsequent cases, we assume $e_T - e_C = 1$, hence providing a lower bound for the persuasion rate.

percent greater than the purchases of the control group, but this difference is not statistically significant.³ Bertrand et al. (forthcoming) randomize the content of direct mail solicitations sent to customers of a South African lender. They vary the interest rates offered, as well as “persuasive” features of the mailer such as the picture displayed or the number of example loans presented. Some features of the mailers—the picture displayed, for example—do have large effects on loan takeup while others do not—comparisons with competitors, for example.

A final body of evidence on advertising effects exploits naturally occurring variation. Bagwell (2007) reviews the large literature studying correlations between monthly or annual measures of aggregate advertising intensity and sales. These studies frequently find large advertising effects, although the endogeneity of advertising expenditure makes interpreting the results difficult. A more promising approach is to exploit high-frequency variation in advertising exposure. Thaivanich et al. (2000), for example, analyze data from a company that advertises health care referral services on television. The television ads direct customers to call an 800 number listed on the screen. Under the assumptions that (i) customers who are persuaded by the ad to call the 800 number do so in a short window of time, (ii) short-run variation in calls is uncorrelated with the timing of ads, and (iii) intertemporal substitutability of calls is low, this data can allow valid estimation of advertising effects. The authors find that an additional television ad in an hour time block generates on the order of one additional call. Similar assumptions justify studies by Tellis (1988) and Akerberg (2001) who use high-frequency individual-level supermarket purchase data matched to individual-level advertising exposures. The latter study finds that an exposure to one additional commercial per week has the same effect on purchases of an average inexperienced household as a 10-cent price cut.

³The authors also report results for specifications in which the independent variable is whether or not individuals actually saw ads rather than whether or not they were assigned to the treatment group (i.e. actual treatment rather than intent to treat). These “difference-in-difference” results do show a significant positive effect of advertising.

2.2 Persuading Voters

The second form of persuasion we consider is communication directed at voters. Such communications may come from politicians themselves, interested third parties such as “get-out-the-vote” organizations, or the news media. The relevant behaviors include the decision to vote or not and the party chosen conditional on voting.

Early studies of political persuasion find little evidence of effects on voters’ choice of candidates. In two famous studies, Lazarsfeld, Berelson, and Gaudet (1944) and Berelson, Lazarsfeld, and McPhee (1954) track the voting intentions and political views of survey respondents over the course of presidential campaigns. They find that after months of exposure to intensive political communications, most respondents are more convinced of the political views they began with. Very few switch allegiance from one party to the other, and such switching is if anything negatively related to the extent of exposure to the campaigns. Summarizing this and other survey-based research, Klapper (1960) writes “[R]esearch strongly indicates that persuasive mass communication is... more likely to reinforce the existing opinions of its audience than it is to change such opinions” (50). The consensus that communication had “minimal effects” dominated research in political science, psychology, and communications for decades (Gerber et al. 2007).

More recently, a number of political science studies have identified stronger relationships between exposure to political communications and voting.⁴ In many cases, the variation in communications that identifies these models is strongly correlated with other factors that affect voter behavior, making the results difficult to interpret. We do not review this literature here.

In a rare field experiment studying the effect of political advertising, Gerber et al. (2007) works with a gubernatorial campaign in Texas preparing a major television and radio advertising buy. Rather than roll out the ads in all markets as would typically be the case, the campaign randomly assigns markets to receive the ad campaign relatively early or rel-

⁴See Iyengar and Simon (2000) and Geys (2006) for reviews.

atively late. Using a daily tracking survey of voters' attitudes, the authors find strong and significant effects of television ads on reported favorability toward the various candidates.

A separate literature examines the impact of get-out-the-vote (GOTV) operations. Among the earliest studies, Gosnell (1926) provided a card reminding of the necessity of registration to a treatment group of 3,000, and no card to a control group of 2,700 voters. This card increased the registration rate from 33% in the control group to 42% in the treatment group, implying a persuasion rate $f = 13.4$ percent. Eldersveld (1956) studied the efficacy of different GOTV methods and found very large effects of GOTV campaigns, especially for personal contacts (door-to-door and phone). However, the sample size of 391 voters is too small to make precise inferences.

Building on these early studies, Alan Gerber, Donald Green, and co-authors run a series of GOTV field experiments (see Green and Gerber, 2008 for a quadrennial review of this literature). In most of these experiments, households within a precinct are randomly selected to receive different GOTV treatments right before an election; the effects are measured using the public records of turnout which can be matched ex post to the households targeted. This methodology avoids the biases associated with self-reports of voting behavior. Gerber and Green (2000) assign GOTV messages to 29,380 voters in a 1998 New Haven congressional election. Door-to-door canvassing raises turnout from 44.8 percent to 47.2 percent which, taking into account that only 27 percent of households in the treatment group are contacted, implies a persuasion rate of $f = 15.6$ percent. Mailers of 1-3 cards with the same content have a much smaller effect, implying a persuasion rate of only $f = 1.0$ percent. Green, Gerber, and Nickerson (2008) find large effects of door-to-door campaigning in 2001 local elections (persuasion rate of $f = 11.8$ percent), and Green and Gerber (2001) demonstrate large effects of personal phone calls in the 2000 election (persuasion rate of $f = 20.5$ percent in one sample and $f = 4.5$ percent in another). (The effect of phone calls from marketing firms is instead much smaller) The remarkable effectiveness of these simple interventions with a personal contact (often a single phone call or personal visit) indicates how malleable

the turnout decision is. A caveat is that these high persuasion rates apply to local election where voters are likely to be less committed; convincing voters to turn out in presidential elections is likely to be harder.

Beyond the literature on direct political communication, a number of recent papers focus on the effect of the news media. Four papers consider the role of partisan media on vote share with different identification methods: two papers exploit idiosyncrasies of media technology that lead to variation in availability, one paper uses the precise timing of media messages, and a last paper uses a field experiment. In the first category, DellaVigna and Kaplan (2007) study the effect of the introduction of the Fox News channel on voting in the 2000 election. They show that whether towns had Fox News added to their cable systems between 1996 and 2000 is largely idiosyncratic, conditional on a set of controls. Using the voting data for 9,256 towns, they find that the Presidential vote share for Republicans in 2000 is half-a-percentage point higher in the towns offering Fox News, implying a persuasion rate of $f = 11.6$ percent of the audience that was not already Republican⁵. They find a similar effect for Senate races, implying that the effect of Fox News extends also to races they do not cover. Enikolopov, Petrova, and Zhuravskaya (2009) use a related method to study persuasive effects of media in Russia. In the 1999 parliamentary elections, all major television networks strongly supported the government party except for a single independent station (NTV). NTV was only available in some parts of the country, and the authors argue that this variation was largely driven by idiosyncrasies of the network. The authors show that in areas with NTV access more voters supported the anti-government party, implying a persuasion rate of $f = 10.2$ percent. Knight and Chiang (2008) study the effect of newspaper endorsements on reported attitudes of voters, exploiting the fact that different newspapers release their endorsements at different times. They detect significant effects of endorsements in cases where the endorsement was a departure from the expected endorsement policy of the newspapers, but no effect in situations where it was not. Finally, Gerber et al. (2009) uses a field experiment to estimate the effect

⁵The persuasion rate for vote share as outcome variable acknowledges that the message can cause party switches or turnout changes. See Appendix A for the expression.

of partisan media coverage. They randomly assign subscriptions to a right-wing newspaper (Washington Times) or a left-wing newspaper (Washington Post) in northeastern Virginia one month before the November 2005 Virginia gubernatorial election. The authors use a post-election survey to measure voting behavior, information, and attitudes after the election. On most measures, they do not detect systematically significant effects of the intervention. They do find a substantial increase in the share of (self-reported) Democratic voters for exposure to the left-wing newspaper, corresponding to a persuasion rate of $f = 20.0$ percent.⁶ Overall, these papers indicate a fairly consistent persuasion effect of 10 to 20 percent for news coverage by a partisan media. While at first look the point estimates of the effect on vote share appear very different (0.4 pp. for Fox News, 2.2 pp. for NTV, 11.2 pp. for The Washington Post), these differences are reconciled by taking into account the exposure rates.

A second set of paper of papers on media effects consider the impact of the media on voter participation. Gentzkow (2006) uses a natural experiment—the diffusion of television in the 1950’s—to study the effects of access to television on voter turnout. Although the timing of television’s introduction was not random, the area reached by stations from a given city is large. He is therefore able to compare observably similar counties that received television relatively early or relatively late. He finds that the introduction of television is associated with a drop in voter turnout of 2 percentage points per decade for congressional elections, corresponding to a persuasion rate $f = 4.3$ percent. He documents that the decrease in turnout is largely attributable to the substitution of local newspapers for TV, leading to less knowledge about local politics. Consistently, the turnout drop is much smaller for Presidential elections which are well covered by the television. Gentzkow and Shapiro (2009) exploit sharp timing to estimate the effect of newspaper entries and exits on voting using data from 1868-1940. They find that an additional newspaper significantly increases voter turnout by .8 percent, implying a persuasion rate $f = 5.1$ percent. Both the qualitative

⁶However, they also find that the share of Democratic voters increases (insignificantly) after exposure to the right-wing paper, possibly because both newspapers reported information suggesting that the Democratic candidate was better than expected.

finding and the size of the persuasion effect are consistent with the findings on introduction of television. They also find that partisan newspapers do not have significant effects on the share of votes going to Republicans, differently from the other studies surveyed above.

2.3 Persuading Donors

The third form of persuasion we discuss is communication that solicits contributions to non-profits or charities. This literature has focused mainly on direct communications from the relevant non-profit. The outcome measures are monetary donations or non-monetary contributions, such as of time or blood.

Some of the earliest studies are in psychology and sociology and examine the determinants of blood donation, recycling, and helping decisions, often using experiments. Among the other determinants, these studies document the importance of in-person contacts (Jason et al., 1984) and the delicate role of incentives in inducing pro-social behavior: for example, paying people to donate blood may reduce the donations because it crowds out intrinsic motivation (Titmuss, 1971). Recently, two economists (Mellstrom and Johannesson, 2008) test this hypothesis using a field experiment. Among women, offering a monetary compensation of about \$7 lowers the rate of blood donations from 51 percent in the control group (with no pay) to 30 percent. Consistent with the intrinsic motivation hypothesis, there is no decrease if the money can be donated to charity. There is no effect on men.

Turning to the studies of fund-raising and monetary donations, academics only recently started studying the effectiveness of different forms of solicitation.⁷ List and Lucking-Reiley (2002) send letters to raise funds for the purchase of computers for a center at the University of Central Florida. Importantly, the amount of seed money (the amount already raised) stated in the different letters is randomized. The main result is that in the low-seed treatment, 3.7 percent of recipients donate a positive amount, compared to 8.2 percent in the

⁷There is an extensive literature examining the response of charitable donations to variables such as tax rates and income shocks (for a review, see Andreoni, 2006). This literature, however, typically relies on aggregate measures of donations to charity from tax filings and does not examine the interaction of the individual with a fund-raiser.

high-seed treatment. This substantial effect of seed money on giving is likely due to inference of higher quality of the charity in presence of high seed money. This paper spawned a very active literature on fund-raising studies: charities collaborate with researchers and set up field experiments to investigate the impact of different fund-raising variables, such as the personal content of the solicitation (in-person versus via mail) and the effect of gifts.

Landry et al. (2006) conduct a door-to-door campaign to fund-raise for a capital campaign at East Carolina University. In their benchmark treatment (Table 1), 36.3 percent of households open the door and 10.8 percent of all households contacted (including the ones that do not open the door) donate, for an implied persuasion rate $f = 29.7$ percent. DellaVigna, List and Malmendier (2009), also using a door-to-door field experiment, estimate the willingness of households to complete a 10-minute survey on charitable giving (for no pay). Among the households that answer the door, 24.8 percent complete the survey. Interestingly, the persuasion rate for a donation or for survey completion in a door-to-door campaign is very similar to the persuasion rate for get-out-the-vote door-to-door campaigns, suggesting that consumers are very responsive to personal solicitations (Table 1). Indeed, in another treatment of DellaVigna et al. (2009), as many as 10.9 percent of households opening the door donate to an out-of-state charity, despite the fact that this charity is consistently ranked last in a series of 5 charities in the survey treatment. This suggests that a genuine desire to give is not necessarily the reason for donations in fund-raising drives.

In another example, Falk (2007) examines the importance of small gifts in triggering higher donation. The context is the mailing of 9,846 solicitation letters in Switzerland to raise money for schools in Bangladesh. One third of the recipients receives one postcard designed by the students of the school, another third receives four such postcards, and the remaining third receives no postcards. The three mailings are otherwise identical. The donations are increasing in the size of the gifts. Compared to the 12.2 percent frequency of donation in the control group, the frequency is 14.4 percent in the small gift and 20.6 percent in the large gift treatment. The large treatment effects do not appear to affect the donations

at next year’s solicitation letter, when no gift is sent. Gifts, therefore, trigger substantial positive reciprocity.

2.4 Persuading Investors

The fourth setting that we consider is communication directed at investors. A first form of this communication is the release of earnings announcements and other financial statements by companies themselves. A second is the release of recommendations and forecasts by financial analysts. In both cases, because individual data on investor response is typically not available, the studies usually focus on aggregate response of stock prices, using an event-study methodology. As such, these studies capture the response of the marginal, as opposed to individual, investor.

The first event study examining investor response to earnings information is Ball and Brown (1968). Using the difference between current and previous annual earnings as a measure of new information, the authors find a significant correlation with stock returns in the two days surrounding the earnings release. To get at more precise measures of the new information, most recent studies use the median analyst forecast (the so-called consensus forecast) to model investor expectations (see Kothari, 2001 for a review). The response to earnings news using this measure is substantial. Companies in the top decile of earnings news experience on average a positive return of 3 percent in the two days surrounding the release, compared to an average -3 percent return for companies in the bottom decile of news.⁸ This information is incorporated very quickly: using a sample of time-stamped announcements, Lee (1992) shows that most of the stock price response occurs within 30 minutes of the announcement.

This evidence is consistent with efficient information processing by the receivers, in line with the underlying incentives and the presence of arbitrage. However, a phenomenon first captured by Ball and Brown (1968) and later studied among others by Bernard and Thomas

⁸These numbers, which are from DellaVigna and Pollet (2009) are representative of the literature.

(1989), does not readily fit into this schema. Companies which release positive earnings news experience continuous increases in stock prices over the next 3-4 quarters, with the opposite pattern happening for companies releasing negative news. This phenomenon, called the *post-earnings announcement drift*, is puzzling to the theory of efficient markets since predictable stock returns should be arbitrated away.

An important third party in the company-investor communication is the analyst. Since collecting information on the profitability of a company involves significant costs, analysts specialize in covering few stocks (5-25 typically) within an industry and provide the information to investors. In this respect, the function of analysts in financial communications is similar to the one of the media in political reporting. Analysts provide investors with three types of information about the performance of a traded company: (i) stock recommendations—a buying or selling advice (Strong Buy/Buy/Hold/Sell/Strong Sell); (ii) earnings forecasts—the forecasted profit per share at some future quarterly earnings announcement (e.g., 5 cents per share); (iii) long-term growth estimates—the forecast for the growth of earnings in the long-term (e.g., 3 percent per year). These pieces of information are observable by investors soon after release and are attentively monitored.

An early study of the impact of analyst recommendations is Cowles (1933). Using data on 7,500 stock recommendations made by analysts in 16 leading financial services companies between 1928 and 1932, he estimates that the mean annual returns of following the recommendations of the analysts is negative at -1.43%. Little research follows this negative result on the recommendation of analysts until systematic data sets of analyst recommendations and forecasts become available in the 1980s and 1990s. Womack (1996), using one of these data sets (First Call), shows that on the three days surrounding the addition of a stock to a list of Buy recommendations stock returns increase 3.27 percent; conversely, the addition to a list of Sell recommendations is accompanied by negative -4.32 percent returns. These returns are about the same size as the response to very positive or very negative earnings news (see above) and are not themselves explained by earnings news. A second key finding

of this study is that, as for earnings announcements, there is a positive drift for new Buy recommendations and a negative drift for new Sell recommendation in the 6 months following the addition. Later studies such as Barber, Lehavy, McNichols, and Trueman (2003) largely confirm these results.

3 What Models Best Capture the Response to Persuasive Communication?

The evidence reviewed above demonstrates the effectiveness of persuasive messages. What factors determine the way a particular message affects receivers' behavior? Existing models of persuasion effects can be divided broadly into two categories.

In the first category, persuasion affects behavior because it changes receivers' beliefs. This includes models where receivers are rational Bayesians, such as informative (Stigler 1961; Telser 1964) and signalling (Nelson 1970) models of advertising, cheap talk models (Crawford and Sobel 1982), persuasion games (Milgrom and Roberts 1986), and persuasion mechanisms (Kamenica and Gentzkow 2009). It also includes models where receivers are not fully Bayesian, and think categorically (Mullainathan 2001; Fryer and Jackson 2003; Mullainathan, Schwarzstein, and Shleifer 2008), have limited memory (Mullainathan 2002; Shapiro 2006), double-count repeated information (DeMarzo, Vayanos, and Zwiebel, 2003), or neglect the incentives of the sender (Eyster and Rabin 2009). We refer to these models as "belief-based."

In the second category, persuasion affects behavior independently of beliefs. This includes models such as Stigler and Becker (1977) and Becker and Murphy (1993) where advertising enters the utility function directly, as well as older models of "persuasive" advertising (Braithwaite 1928). This category also includes some dimensions of psychological models of persuasion where non-informative "peripheral" factors may play a central role (see e.g. Petty and Cacioppo 1986; Petty and Cacioppo 1996). We will stretch terminology somewhat

and refer to all such models as “preference-based.”

Distinguishing different models of persuasion is particularly important because they have very different implications for welfare. In a simple belief-based model with rational receivers, the existence of persuasive communication cannot make receivers worse off. In this case, increasing the supply of information and making information markets more competitive will tend to improve welfare. Welfare effects are harder to evaluate if receivers are not Bayesian or if persuasion works through emotion or preference change.

Regardless of the model, messages are only effective if they are actually received. In many settings, receivers make active decisions about what information to seek out and how to allocate their attention, and the way they make these decisions has important implications for the effectiveness of persuasion in equilibrium. At the end of this section, we briefly review evidence on the way endogenous exposure mediates persuasion effects.

3.1 Predictions of Belief-Based Models

A first core prediction of most belief-based models is that persuasion will tend to be more effective when receivers are less certain about the truth. In Bayesian models, the weaker are receivers’ priors, the more their beliefs are affected by a given piece of new information. This may (though need not) imply larger effects on behavior. In limited memory or categorical thinking models, behavior will also typically be less elastic when receivers are close to certain about the state *ex ante*.

Akerberg (2001 and 2003) tests the prediction that advertising has a larger effect for consumers who have little experience with a product. His study combines data on a panel of consumers’ purchases with data on their exposure to television advertisements, focusing on a new brand of yogurt Yoplait-150. Identification of advertising effects here is non-experimental, and relies on individuals’ exposure to ads for the yogurt being orthogonal to their pre-determined taste for yogurt. This assumption would be violated, for example, if advertisers run ads on programs whose viewers are differentially likely to buy yogurt.

Akerberg shows that consumers who have bought the product in the past (and so have more information about its characteristics) are less sensitive to advertising than those who have never bought, consistent with the prediction. As mentioned earlier, the effect of an additional ad per week on inexperienced households is equivalent to a 10 cent price cut; the effect on experienced households is a fifth the size and not significantly different from zero.

Several of the other studies summarized in Section 2.1 also look at the interaction between advertising and consumer experience. Simester et al. (2007) shows that the effect on purchases of sending catalogs is larger on the intensive margin for consumers that previously purchased infrequently than for consumers that previously purchased frequently. The first group is less likely to have precise priors on the products offered. Thaivanich et al. (2000) find that advertisements for the health care referral service have larger effects in markets where the service was only recently introduced. A number of studies beginning with Shryer (1912) find diminishing returns to repeating the same advertisement (at least as the number of repetitions gets large), which is consistent with receivers' priors becoming increasingly strong over time.

The role of receiver priors has also been tested in the domain of political persuasion. Zaller (1992) shows that individuals with high political awareness updated their views of Ronald Reagan less in the wake of the Iran-Contra affair, in part because their prior knowledge of Reagan's performance was more extensive. Enikolopov, Petrova, and Zhuravskaya (2009) find that the effect of access to independent television news in Russia on voting is largest for those with less prior political knowledge and those who use alternative information sources less. Prior (2006) uses survey data from 1958-1970 to estimate the cross-sectional correlation of knowledge of incumbent House politicians and propensity to vote for incumbents on the one hand, and availability of television stations on the other. He finds no systematic relationship on average, but finds a positive correlation for less-educated voters. Such correlations must be interpreted with caution, but they are consistent with the view that the effects of television should be greatest among voters who have the least prior knowledge.

A second prediction specific to Bayesian belief-based models is that receivers' inferences from a given message will depend on what they know about the credibility of the sender. If a university changes its standards so that a larger share of students receive A's, the impact of knowing a student received an A on a receiver's beliefs should become smaller. A republican endorsement from a newspaper that always endorses republicans should carry less weight than a republican endorsement from a newspaper that typically endorses democrats. Non-Bayesian models do not necessarily share this prediction, and Eyster and Rabin (2009) posit specifically that receivers will adjust too little for sender credibility.

The study by Knight and Chiang (2008) discussed above provides some evidence from the political arena. Consistent with the Bayesian model, they find large effects of newspaper endorsements on voting only in cases where the endorsement is a "surprise" in the sense that the paper supports a party it typically opposes. When a newspaper endorses the same party repeatedly, their endorsements have no significant effect.

Most evidence on source credibility comes from the literature on financial analysts. Kadan et al. (2008) study a 2002 regulation that required brokerage houses to publicize the share of their analysts' recommendations in the sell, hold, and buy categories (NASD Rule 2711 and amended NASD Rule 472). In response, most houses increased the share of sell and hold recommendations: after the regulation, about 20 percent of the recommendations are sell and 44 percent hold, up from, respectively, 5 percent and 26 percent before the regulation. Consistent with the Bayesian model, stock returns after the reform respond more to buy recommendations (which are now less common and hence more informative) and less to sell and hold recommendations (which are more common and less informative). Of course this does not show that receivers make correct inferences from these recommendations, only that their inferences shifts in the predicted direction.

Additional evidence on source credibility comes from studies of analysts' conflicts of interest. As discussed at more length below, analysts face strong incentives to provide positive recommendations. These incentives are especially strong for affiliated analysts—those

whose companies also provide investment banking services to the companies the analysts cover. Malmendier and Shanthikumar (2007) ask whether investors take these incentives into account. They first replicate previous findings showing that affiliated analysts systematically give more buy or strong buy recommendations than unaffiliated analysts. The bias is sufficiently strong that a rational investor should essentially neglect the recommendations of affiliated analysts. The authors then show that institutional investors (proxied by large trades) do not respond to buy and strong buy recommendations issued by affiliated analysts, while individual investors (proxied by small trades) appear to take the recommendations at face value: they buy in response to buy recommendations and hold in response to hold recommendations, and do not distinguish between affiliated and not affiliated analysts. De Franco, Lu, and Vasvari (2007) focus on a specific set of analyst recommendations that were alleged by the SEC to have been deliberately distorted as a result of conflicts of interest. Consistent with Malmendier and Shanthikumar (2007), they show that individual investors buy in response to the optimistically-biased recommendations. They also show that institutional investors *sell* in response to the optimistic recommendations, consistent with them recognizing both that the recommendations are biased and that individual investors will over-respond. Taken together, this evidence suggests that sophisticated agents adjust for sender credibility as the Bayesian model would predict, while inexperienced agents under-adjust as in Eyster and Rabin (2009).

We discuss one laboratory study in this connection because it is so directly relevant to the issue at hand. Cain, Loewenstein, and Moore (2005) conduct a simple experiment where subjects are paid for the precision of the estimates of the number of coins in a jar. Since they see the jar only from a distance, they have to rely on the advice of a second group of subjects, the advisors, that inspect the jar from close. The two experimental treatments vary the incentives for the advisors. In a first treatment, the advisors are paid for how closely the subjects guess the number of coins; in a second treatment, the advisors are paid for how high the subjects' guess is. Despite the fact that the incentives are common-knowledge,

the estimate of the subjects is 28 percent higher in the second treatment. The leading interpretation of this finding is that the subjects do not discount enough for the conflict of incentives of the advisors.

3.2 Predictions of Preference-Based Models

The simplest prediction that distinguishes preference-based models from belief-based models is that the content of messages may affect behavior even when it conveys no information. Conclusively identifying such content is difficult, because even apparently non-informative features of messages may still serve as informative signals in equilibrium (Nelson 1970). Also, non-informative dimensions may affect belief formation through channels such as framing, salience, and attention (Zaller 1992; Mullainathan, Schwarzstein, and Shleifer 2008), making the distinction between belief-based and preference-based models blurry. Nevertheless, many pieces of evidence on how persuasion operates are hard to rationalize with at least the simplest belief-based models of persuasion.

One piece of evidence is the content of messages themselves. Many have noted, for example, that advertisements typically contain imagery, humor, narrative, and other content that does not convey any obvious information. Resnik and Stern (1977) estimate that only half of broadcast television advertisements can be classified as informative. Mullainathan and Shleifer (2005) note the example of a shampoo brand advertised with a slogan “We put silk in a bottle,” even though the firm acknowledges that silk has no effect on the performance of shampoo. Mullainathan and Shleifer (2005) also show that the non-informational content of Merrill Lynch advertising campaigns varies systematically with market conditions. Their ads’ images evoke security and reliability (e.g. a grandfather fishing with his granddaughter) when market growth is low, and evoke modernity and aggressiveness (e.g. a bull wired to a circuit board) when market growth is high. They argue that these patterns are most consistent with a model where firms design ads to tap into consumers’ existing beliefs rather than convey or signal new information about their products.

In a demonstration of the effect of non-informational content, Bertrand et al. (forthcoming) randomize the content of direct mail solicitations sent to customers of a South African lender. They vary the interest rates offered, as well as features of the mailer such as the picture displayed and the number of “example loans” presented. The authors find that the persuasive features are jointly (marginally) significant in predicting loan takeup. The manipulations with the strongest effects are: (i) presenting a simple table with only one example loan rather than several; (ii) not including a suggested use for the loan; (iii) and including a picture of an attractive female. All three of these effects are roughly equivalent to a two percentage point reduction in the monthly interest rates (Table 1, the rate varied from 3.25 percent to 11.75 percent). The pattern of results is not clearly organized by any particular theory of how advertising content matters, but it provides evidence that non-informational dimensions such as the attractive picture can have economically large effects.

Landry et al. (2006) present related evidence from the door-to-door fund-raising campaign discussed in section 2.3. They show that a one standard deviation increase in attractiveness of the female solicitors increases the amount raised by 50 percent. The effect only occurs when males open the door. This suggests that the personal interaction impacts the willingness to pay. The importance of personal interaction is also consistent with the pattern of results in Gerber and Green (2000) who find the largest persuasion rates for in-person visits, followed by phone calls and then mailers.

A second prediction that distinguishes preference-based models is that receivers may take costly steps to avoid exposing themselves to persuasion. Because there is free disposal of information, messages in a belief-based model cannot make receivers worse off in expectation (except for a time cost), so we should not generally see costly avoidance behavior. Becker (1996) notes the example of pedestrians who cross the street to avoid passing a beggar. This behavior is hard to understand unless passing the beggar, or especially passing the beggar and refusing to give money, has a direct negative effect on utility. DellaVigna, List, and Malmendier (2009) test directly for costly avoidance in the context of charitable giving using

a field experiment. They compare the fund-raising in a standard door-to-door campaign to a campaign in which the households are notified a day in advance about the upcoming fund-raising drive with a door-hanger. They find that the door-hanger reduces the share of households at home by 10 to 25 percent and lowers giving by 30 percent if the door-hanger has a ‘Do Not Disturb’ check-box. This suggests that the persuasion of donors operate in part through social pressure, inducing households to take costly steps to avoid exposure to the solicitation.

3.3 Demand for Information and Limited Attention

The effectiveness of persuasion depends in many settings on receivers’ decisions about what information to seek out and how to allocate attention.

Several studies show that potentially valuable information is ignored even when it is public information, due to either high costs of information acquisition or non-rational limited attention (see DellaVigna 2009). Huberman and Regev (2001) chronicle the market reaction to results suggesting a drug patented by the company EntreMed might cure a type of cancer. On November 28, 1997, these results are reported by *Nature* and on page A28 by the *New York Times*. Unsurprisingly, the stock price of EntreMed increases by 28 percent. Five months later, on May 4, 1998, the New York Times publishes a front page an article on EntreMed that is almost identical to the article published earlier. Despite the fact that the article contains no new hard information, it leads to a 330 percent one-day return for EntreMed, and to a 7.5 percent one-day return for all bio-tech companies, moving billions of dollars in market capitalization. The stock price of EntreMed does not revert to the previous level over the whole next year. Several interpretations of these events are possible, but what is clear is that the way public financial information is disseminated can significantly affect the market reaction.

Two studies document more systematically the importance of limited attention in the context of quarterly earnings releases by companies. As we discussed in Section 2.4, com-

panies that release positive earnings news experience an increase in stock returns over the next 2-4 quarters. A leading interpretation of this phenomenon is that investors are slow to respond to the new pieces of information (Hong and Stein, 1999). If this is the case, one would expect even more underreaction to earnings news on days on which investors' costs of processing the information are high. DellaVigna and Pollet (2009) examine releases taking place on Friday (versus other weekdays), under the assumption that the weekend is a distracting force. Hirshleifer, Lim, and Teoh (forthcoming) examine releases taking place on days with a higher number of earnings announcements, under the assumption of attention as a fixed resource. The two papers find very similar evidence that on low-attention days the initial incorporation of the earnings news into stock returns is 20 percent slower, with a higher drift. An obvious confound for these papers is the fact that companies can time the release of their earnings and therefore the companies that release on low-attention days may systematically differ. However, the selection would need to explain why companies releasing on low-attention days experience both less immediate response and more delayed response.

The endogeneity of receivers' attention is also important in political persuasion. A persuasive argument to vote Republican will have little effect if it only reaches those who are already committed Republicans. (It can still have an effect if it affects the turnout decision). The classic studies of political campaigns by Lazarsfeld, Berelson, and Gaudet (1944) and Berelson, Lazarsfeld, and McPhee (1954) demonstrate that most voters are exposed mainly to campaign messages consistent with their pre-existing beliefs. Similarly, several studies of news markets document a strong demand on the part of consumers for news that has a political slant consistent with their own views (Gentzkow and Shapiro forthcoming; Gentzkow and Shapiro 2004). This may be driven by both irrational "confirmatory bias" (Lord, Ross, and Lepper 1979) and rational inferences about quality by consumers with heterogeneous priors (Gentzkow and Shapiro 2006). Regardless of the mechanism, it implies that the overall effect of persuasive messages (or of important new information) could be smaller than it would be in a world where receivers' demand for news was independent of their prior beliefs.

4 What are Senders' Incentives?

Most of the studies reviewed so far treat the content or quantity of persuasive messages as exogenous and study the effect on receivers' behavior. In markets, however, the supply of persuasion is endogenous. It is therefore important to understand the forces that determine the supply of persuasive messages in equilibrium.

Senders' incentives in persuasive markets are complex. Their interest in affecting receiver behavior may be direct, as with a politician trying to get elected, or indirect, as with a media firm that wishes to curry favor with a politician. This will naturally give senders an incentive to distort information. A countervailing force for accuracy is the desire to build a reputation: if receivers are rational, senders may benefit from committing to limit the incentive to distort, or to report accurately. These two forces—the incentive to distort and the incentive to establish credibility—may play out differently in different markets, and their relative strength will be a key determinant of the extent to which persuasive communications have beneficial or harmful effects.

Even if a monopoly sender has no incentives to build credibility, a sender in a competitive market may. A media firm that distorts news, for example, may pay a significant cost if this fact is revealed by its competitors, as long as the audience cares sufficiently about accuracy and consumers have largely rational beliefs.

4.1 Incentives of Advertisers

In the simplest model of advertising, firms do not internalize any benefits from accuracy. They just want to sell. As such, communication can be pure cheap talk. This tendency may be limited if senders include verifiable information or if regulation or reputation create incentives for accuracy. Also, advertising campaigns can have an effect even if they do not convey credible information as long as they have a direct impact on preferences. The large scope for distortion in advertising may partly explain the failure to find consistent effects on

receiver behavior (section 2.1).

One domain where distortions in advertising have been quantified is pharmaceutical advertising in medical journals. Villanueva et al. (2003), for example, study 264 advertisements for antihypertensive and lipid-lowering drugs published in Spanish medical journals in 1997. They count the number of claims in the ads that were supported by bibliographic citations, and then assess whether the studies cited indeed back up the claims. In 44 percent of cases, the authors find that the claims are not supported by the studies. Cooper and Schriger (2005), van Winkelen et al. (2006), and Greving et al. (2007) perform similar analyses, finding large numbers of claims that are not supported by research or that are supported only by research sponsored by the manufacturer.

The conflict of interest induced by the advertisers' incentives implies an important role for potentially objective third parties. *Consumer Reports*, for example, provides product ratings, *US News and World Report* provides university rankings, and the *American Democracy in Action* provides quantitative measures of the voting record of politicians. The information from certifiers can have substantial effects on consumer behavior. Jin and Leslie (2003) study the case of the introduction of health scores for restaurants in the L.A. area. Following a prominent food poisoning episode, different areas in LA required restaurants to prominently post a hygiene score. Jin and Leslie find that the introduction of the report cards substantially increases the attendance to highly-rated restaurants and lowers the incidence of food poisoning episodes at the hospital.

4.2 Incentives in Financial Reporting

Firms often have an incentive to distort their financial reports to make outcomes look more favorable to investors. Such distortions are substantially constrained, however, by regulation. In both quarterly and annual releases, the companies report earnings (profitability) per share, which by regulation must conform to one of two accounting standards (GAAP or “street”). Hence, while the CEO in the periodic communication can emphasize underlying trends,

new products, or significant investments, the attention of the investors is largely focused on one standardized number. Moreover, earnings releases occur at a pre-scheduled time, and firm managers cannot release any profitability information in advance of this call. The high information content of financial releases is consistent with the substantial reaction of investors (section 2.4).

Despite the constraints imposed by regulation, several studies present evidence that firms manipulate reported earnings. DeGeorge, Patel, and Zeckhauser (1999) document an asymmetry in the distribution of earnings surprises: there are substantially more announcements that meet the analyst consensus forecast by 1 or 2 cents than announcements that fail to meet the forecast by 1 or 2 cents. This suggests that firms that would barely miss the forecast, anticipating a negative response by investors, adopt accounting management to be able to meet the expectations.

Hirshleifer, Hou, Teoh, and Zhang (2004) show that firms with a high value for net operating assets, a proxy for accounting management, experience in later quarters worse earnings news, as well as substantially lower stock returns. Accounting management typically involves intertemporal substitution of certain discretionary items, such as accounts receivable, and so are likely to lead to more negative earnings releases in the future. A trading strategy based on the results of this study earns annual returns in excess of 15 percent. This suggests that investors do not fully appreciate the extent of the accounting manipulations, which in turn makes it worthwhile for companies to undertake them.

A more subtle manipulation exploits the two measures of earnings, the GAAP number and the “street” (also called pro-forma) number, which differ in the treatment of extraordinary items such as restructuring charges. Dyck and Zingales (2003) show that firms report the more favorable number more prominently in their press release. They also show that the media coverage of the earnings tend to follow the same pattern. In response, the stock returns in the two days surrounding the announcement responds more to the number that the firm emphasized than to the other number. Again, the fact that investors do not undo the

accounting distortion makes it profitable for firms to engage in such behavior in equilibrium.

4.3 Incentives of Financial Analysts

Unlike companies advertising their products, analysts provide information whose accuracy is easy to measure: earnings forecasts can be compared to the actual release, and recommendations, at least in principle, can be benchmarked against the stock performance. Such ex post verifiability can significantly limit the incentives for distortion. We might expect that maximizing forecast precision should will be the key incentive for analysts.

However, analysts also face strong incentives to provide forecasts favorable to the companies they cover. Analysts that provide positive coverage of a company have easier access to company information and are not subject to continuous pressure by the company to change their outlook. Additional distortions exist for sell-side analysts, that is, the ones that work within investment banks. Investment banks derive little if any revenue from selling the analyst reports which quickly enter the public domain. The major sources of profits are corporate finance transactions, such as mergers and secondary equity issuances. Analysts are therefore under enormous pressure to provide positive coverage of companies that provide such transactions. The analysts covering such companies, affiliated analysts, therefore face a stronger conflict of interest.

Since analysts careers are observable, it is possible to study how promotions and demotions depend on the precision in forecasts and on the bias of the forecasts. Hong and Kubik (2003) measure movements of analysts between high-status brokerage houses (such as Goldman Sachs) and low-status brokerage houses (typically more specialized ones). They relate the likelihood of these movements to measures of the accuracy of forecasts (the mean absolute forecast error for past earnings forecasts, relative to other analysts) as well as measures of optimism (the fraction of forecasts that are above the consensus forecast). The key result is that optimism counts more than accuracy in determining career shifts. An analyst in the top 10 percent of optimism score is 90 percent more likely to be promoted, compared

to the remaining analysts. The corresponding effect of being in the top 10 percent of accuracy is only a 41 percent increase in the promotion probability. Similarly, forecast optimism lowers the probability of demotion more than accuracy. The authors also find (marginally) statistically significant evidence that for demotions of affiliated analysts, who face a stronger conflict of interest, optimism matters more and accuracy less. This evidence implies that the analysts face a very severe conflict of interests.

Does this conflict of interest affect recommendations in equilibrium? One test is to compare the distribution of recommendations to what we would expect if the labels “buy,” “hold,” and “sell” were interpreted literally. In that case, unbiased analysts should on average issue approximately as many recommendations to buy a stock as to sell a stock, while distorted analysts should issue more buy recommendations. Figure 1 provides evidence on the distribution of recommendations (the data is from Malmendier and Shanthikumar, 2007): less than 5 percent of recommendations are Sell or Strong Sell recommendations, and about 60 percent of recommendations are Buy or Strong Buy. The bias in favor of Buy recommendations is even stronger for the affiliated analysts, for which the share of Buy or Strong Buy recommendations is larger than 70 percent.⁹

The size of the distortion is such that on average a sophisticated investor should interpret a buy recommendation as a recommendation to hold a stock, and a hold recommendation as an invitation to sell. In addition, investors should essentially neglect the recommendations of affiliated analysts. As we discussed above, this is precisely how institutional investors appear to respond to recommendations, while individual investors fail to adjust for these distortions (Malmendier and Shantikumar, 2007).

The distortions in quarterly earnings forecast take a more nuanced format. Companies generally prefer optimistic earnings forecasts. However, when the earnings release is imminent, the preference shifts towards cautious forecasts, which are easier to meet or ex-

⁹While the distortions are plausibly due to conflict of interest, an alternative interpretation is that they are instead due to optimism: analysts choose to cover companies that they have a positive outlook for. This alternative interpretation however does not explain why affiliated analysts, who face a stronger conflict of interest, have an even larger bias.

ceed. This is particularly true given that, especially in recent years, top managers can only sell stock in their company in a short time window after the earnings announcement, and they benefit from a positive earnings surprise. Consistent with this, Richardson, Teoh, and Wysocki (2004) document a ‘walk-down’ of analyst forecasts for the 1990s: forecasts are the most optimistic at horizons of one year or more before the release, they become monotonically less optimistic but remain largely above the (ex-post) realized earnings until 3-4 months before the release, at which point they turn pessimistic. In the earlier years (1980s), when manager incentives to boost short-term stock prices are lower, there is still a walk-down, but the forecasts remain optimistic over the whole sample.

Further, we expect the most biased analysts to provide both the most positive recommendations and the least positive earnings forecasts near the release date. Indeed, there is a negative correlation between recommendations and forecasts for affiliated analysts, for whom the conflict of interest is strongest (Malmendier and Shantikumar, 2009).

Analysts forecasts are the only domain we are aware of where there is direct evidence on the effect of competition. Hong and Kacperczyk (2009) consider the shifts in analyst competition induced by the merger of investment banks. Following such mergers, analysts covering similar stocks in the two banks are often laid off. Using these mergers as instruments for decreased competition, they estimate that the remaining analysts that now face less competition increase their optimism bias in earnings forecasts by about 10 percent. This increase in bias holds also when the authors focus on the optimism of analysts working in banks not affected by the merger. The authors also document a similar effect on optimism in the long-term growth forecast and a smaller effect on recommendations. Hence, it appears that competition may moderate the analyst bias, though it is unlikely to erase it.

4.4 Incentives of the Media

Like analysts, media firms face complex incentives. Most obviously, they benefit from increasing consumer willingness-to-pay in terms of money and time for their products. This

generates both direct revenue (newspaper or magazine subscriptions, for example) and revenue from advertisers who are willing to pay more the more consumers they can reach. Whether catering to consumers leads them to provide accurate information depends on what consumers demand. If consumers value accuracy, market incentives will usually push firms to provide it, so long as ex post feedback on which stories were right and which were wrong is sufficiently strong. If consumers do not value accuracy and are just as happy with fiction that entertains them or supports their political beliefs, we expect the scope for distortion to be large.

The scope for distortion also depends on other “supply-side” incentives of firms to manipulate content. Media owners may have personal preferences over political outcomes. Politicians, advertisers, or other interested parties may exert pressure on media firms to mould content in ways these parties prefer. Reporters may prefer writing stories that align with their personal political views.

To evaluate the importance of these forces in the U.S. newspaper markets, Gentzkow and Shapiro (forthcoming) use automated searches to measure the political slant of 420 US newspapers. They use these measures to estimate the demand for slant (the readers’ preferences) and the supply of slant (the newspaper’s preferences). They document a statistically and economically significant demand for slant consistent with consumers’ own ideologies. The data do not reject the hypothesis that newspapers respond to this demand by choosing slant to maximize circulation. The authors then examine directly the role of ownership. Once they control for geographic clustering of ownership groups, they find that two newspapers with the same owner are no more similar politically than two random newspapers, implying small (if any) ownership effects. Also, they do not find any evidence of a role of reporter preferences and the party of local or state politicians in determining slant. They conclude that the role of supply-side incentives in shaping the average slant of U.S. newspapers is likely to be small, and that, on average, the newspaper political slant is determined by demand.

Other studies show that supply side incentives can be important. Larcinese, Puglisi and

Snyder (2007) present a case study of the editorial and news content of the Los Angeles Times before and after Otis Chandler took over in 1960. Chandler sets out to change the previous Republican orientation of the paper. Indeed, the LA Times shifts from endorsing mostly Republican candidates before 1960 to endorsing mostly Democrats by 1970, while no corresponding change in the political preferences of the electorate takes place at the same time. In addition, while before 1960 the journal provided less coverage of high unemployment under Republican presidents than under Democratic presidents, this bias disappears after 1965. Larcinese et al. extend this analysis to a cross-section of newspaper and present some evidence that the partisan coverage of unemployment news responds more to owner preferences (measured by newspaper endorsements) than to local electoral preferences. Gilens and Hertzman (2008) show that newspapers whose parent companies have an interest in television stations affected by a proposed de-regulation of broadcast television cover the debate over that de-regulation significantly differently than those whose parent companies did not have such an interest.

Durante and Knight (2009) test for ownership effects in the Italian media where one entrepreneur, Silvio Berlusconi, owns the three major private television stations and controls several of the national newspapers. Berlusconi also indirectly exercises control over the public television station when his party is in power. Using measures of on-air-time for the prime minister and opposition party leader, Durante and Knight find substantial ownership effects. Berlusconi's television stations devote substantially more time to the prime minister when Berlusconi's coalition is in power than when the other coalition is leading the government. Moreover, the share of time devoted to the different politicians by the public station shift once Berlusconi's coalition comes into power. Even then, one public channel (Rai 3) offer more balanced coverage. Interestingly, some of the audience switches to this channel from the other channels when Berlusconi's party gets into power. This suggests a demand for less politicized information, even though this substitution effect is quite small.

Another form of supply-side incentive comes from advertisers. Since the media depends

heavily on advertising for revenue, advertisers may push quid-pro-quo implicit agreements to increase their advertising expenditure as long as the media coverage is positive towards their products. Reuter and Zitzewitz (2006) show that financial magazines that depend heavily on advertising from mutual funds appear to bias their reviews of funds in favor of their advertisers. The identification is based on considering how the coverage of different mutual funds in the news section (including recommending a fund) varies as a function of the advertising space purchased by a mutual fund family in the previous 12 months. Interestingly, the *New York Times* and *Wall Street Journal* display no such correlation, suggesting that high reputation can indeed limit these distortions.

5 How Does Persuasion Affect Equilibrium Outcomes?

The evidence reviewed so far sheds light separately on the responses of receivers and on the behavior of senders. Often, however, the ultimate welfare effects of persuasion depend on the way senders and receivers interact in equilibrium. In this section, we review a handful of studies that estimate the equilibrium effects of persuasive communications.

One series of four studies considers how communication through the news media affects equilibrium policy outcomes. These are distinguished from the studies reviewed in section 2.2 because they consider the equilibrium response of politicians to persuasive communication between the media and voters. The common theme in these studies is that media coverage leads politicians to respond more to the demands of voters, in most cases plausibly increasing voter welfare. Stromberg (2004) studies the diffusion of radio in the 1930's. Using plausibly exogenous variation in the availability of radio signals, he shows that localities that received radio obtained larger quantities of federal dollars during the new deal, consistent with a model where better information strengthens representatives' incentives.

Similarly, Stromberg and Snyder (2008) ask how representatives whose actions are heavily covered by local newspapers behave differently from those whose actions are less cov-

ered. They find that more newspaper coverage, instrumented with the “congruence” between congressional districts and newspaper markets, leads representatives to exert more effort directed at constituent interests (as measured by testifying before committees and serving on committees that control pork spending), and to bring home more federal dollars.

Besley and Burgess (2002) use data on newspaper circulation, government aid, and economic conditions for a panel of Indian states from 1958-1992 to show that aid responds more to food shortages and floods in areas with higher newspaper circulation per capita (controlling for income and other demographics). They estimate that a 10 percent drop in food production leads to a 1 percent increase in food aid for states at median newspaper circulation per capita, but a 2.3 percent increase in food aid for states at the 75th percentile of newspaper circulation per capita. These results are driven mainly by circulation of local language newspapers.

Finally, Eisensee and Stromberg (2007) look at the way news media coverage influences the provision of foreign aid in response to natural disasters. They first show that disasters that occur at the same time as other newsworthy events, such as the Olympics, receive much less news coverage than disasters that occur at other times. Using this variation to instrument for news coverage of disasters, they estimate that news coverage increases the probability that the US sends aid by about 16 percent on average, and by as much as 70 percent for stories on the margin of being covered or not. They interpret these results to mean that public outcry over disasters is largely driven by news coverage and that it is the public response which provides an incentive to politicians to send aid.

A second category of studies on the equilibrium effects of persuasion is the extensive empirical literature on the indirect effects of advertising. The direct communication between advertisers and consumers in turn affects market structure, profits, product quality, and prices. As Bagwell and others have noted, many of these papers study cross-sectional or time-series correlations between measures of advertising intensity and outcome variables. Given that advertising is simultaneously determined with most of these outcomes, most of

these results do not have a clear causal interpretation. Here, we focus on one strand of this literature that we find most interesting: evidence on the way limits on retail advertising affect equilibrium retail prices. This literature speaks directly to the question of whether advertising appears to promote competition by decreasing search costs and increasing market transparency, as argued by Stigler (1961) and others, consistent with welfare-increasing effects of advertising.

Bagwell (2007) reviews a number of studies beginning with Benham (1972) that look at the cross-sectional relationship between advertising restrictions and price levels. In markets for eyeglasses, prescription drugs, gasoline, and optometry, these studies all find that prices are significantly lower in markets where price advertising is allowed relative to markets where it is banned. Of course, advertising restrictions could be correlated with other determinants of prices.

Two studies use changes in firms' ability to advertise, rather than cross-sectional correlations. Glazer (1981) studies a 1978 newspaper strike in New York City. The strike significantly reduces the supply of newspaper advertising space. Glazer studies the prices at supermarkets in Queens which depend especially heavily on newspapers for price advertising, using supermarkets in neighboring Nassau county (which was much less affected by the strike) as a control. He reports that prices of the six products studied rises about 6 percent in Queens relative to Nassau during the strike and returns to their earlier relative levels after the strike. He also shows that prices in local groceries in Queens (which do little newspaper advertising) do not change significantly.

Milyo and Waldfogel (1999) study a 1996 Supreme Court decision that ended a ban on price advertising by liquor stores in Rhode Island. Consistent with a pro-competitive effect of advertising, the Rhode Island Liquor Stores Association was on record as supporting the advertising ban because they believed "smaller retailers would be devastated" by price advertising by larger (and lower-price) stores. The authors estimate that, relative to a control group of stores in Massachusetts not affected by the ban, prices fell by between 0.4 and 0.8

percent in Rhode Island after the ban, although, in contrast to the earlier literature, these differences are not statistically significant.

6 Conclusion

How does persuasion operate? We have analyzed communication directed at consumers, voters, donors, and investors. The evidence suggests that persuasive communication is more effective for voters, donors and investors than it is for consumers. This could be due to the fact that there is more crowding of messages to consumers. The evidence also suggests that personal contact is a key determinant of the effectiveness of persuasion across contexts. While this may be due to differences in the information conveyed, it is consistent with social pressure playing an important role.

We have also surveyed how the evidence supports different models of persuasion. The predictions of belief-based models with rational receivers clearly find support in the data. At least in some circumstances, receivers respond more when they are more uncertain and when messages are credible, senders are constrained by market forces to limit distortion, and increasing the supply of persuasive messages makes markets more competitive. At the same time, there is also clear evidence of systematic departures from these models. Unsophisticated receivers neglect incentives, non-informative dimensions of messages affect behavior, and receivers take costly steps to avoid persuasive communication.

Testing these models is important in part because of their implications for welfare. The broad conclusion from theory is that welfare effects are more likely to be positive when persuasion functions as information and receivers are rational. The evidence in this survey suggests that the answer will depend on the setting and on the receiver: small and large investors, for example, display very different levels sophistication.

We see three especially important gaps in the existing literature. First, there is little evidence on the long-term effects of persuasion which, for many questions, may be the most

important horizon. The modern empirical literature has made great strides by exploiting both randomized and natural experiments, but these experiments often privilege the estimation of short-term effects. For example, of the fifteen studies reviewed in Table 1, only four examine impacts over a horizon of a year or longer.

Second, few studies are designed to test directly models of persuasion. For example, there is limited evidence on how receivers take into account sender incentives or on how receivers seek information. Further, most of the existing evidence provides qualitative, as opposed to quantitative, tests of the models. This makes it difficult to estimate precise economic and welfare implications. As a starting point, a consistent quantitative measure of persuasion effects, the persuasion rate (Table 1), can resolve apparent inconsistencies across studies.

Third, some topics of great importance have been largely neglected so far. A first one is the role of competition. The proposition that competition among senders will limit the scope for distortion has played a central role in the way speech is regulated in many domains, with very limited evidence to bear. A second one is the role of political persuasion in non-democratic societies. Virtually all the evidence is from the US or other democracies. Yet, one of the original motivations for studying persuasion is its role in autocracies and dictatorships. On all these topics, more evidence is clearly needed.

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Appendix – Documentation of Table 1

[Not for publication]

We discuss here the data used for the construction of the information used to compute persuasion rates in Table 1.

Persuading Consumers. For the Simester et al. (2007) paper, the information comes from personal communication with the authors. We do not observe the exposure rate (how many more people read the catalogs in the treatment group) and hence assume $e_T - e_C = 1$. For the Bertrand et al. (forthcoming) paper, we take the take-up in the control group t_C from the Summary Statistics (Table II), and add the treatment effect (Table IIIa) for respectively a female photo (.0057) and 2 points of interest rate ($2 \times .0029$). Even though the interest rates vary between 3.25% and 11.75%, for illustrative purposes we take the average interest rate for the Medium Risk group (6.5% monthly) as the control group.

Persuading Voters. For Gosnell (1926), we take the estimate of t_T and t_C from the second paragraph on page 870. For Gerber and Green (2000), the results are from Table 3 for the effect of canvassing and from Table 5 for the effect of the mailers. For Green, Gerber, and Nickerson (2003), the estimation of t_T and t_C is based on Table 1 in their paper. The estimate of exposure rate difference comes from the third column in Table 2. For Green and Gerber (2001), we calculate the t_T and t_C and exposure rate difference based on Table 3 in their paper.

For the calculation of effects on vote share, the computation of persuasion effects is slightly different (DellaVigna and Kaplan, 2007). Denote by v_i is the vote share for the party supported by the media in treatment i , t_i is the turnout, and v_0 and t_0 are the respective variables in absence of an intervention. The vote share in treatment j equals $v_j = (r_0 + (1 - r_0) e_j f) / t_j$. Writing out $v_T - v_C$ and substituting for $t_j = t_0 + (1 - t_0) e_j f$ to simplify the expression, one obtains

$$f = \frac{v_T - v_C}{e_T - e_C} \frac{t_C t_T}{t_0 (1 - v_0)}. \quad (2)$$

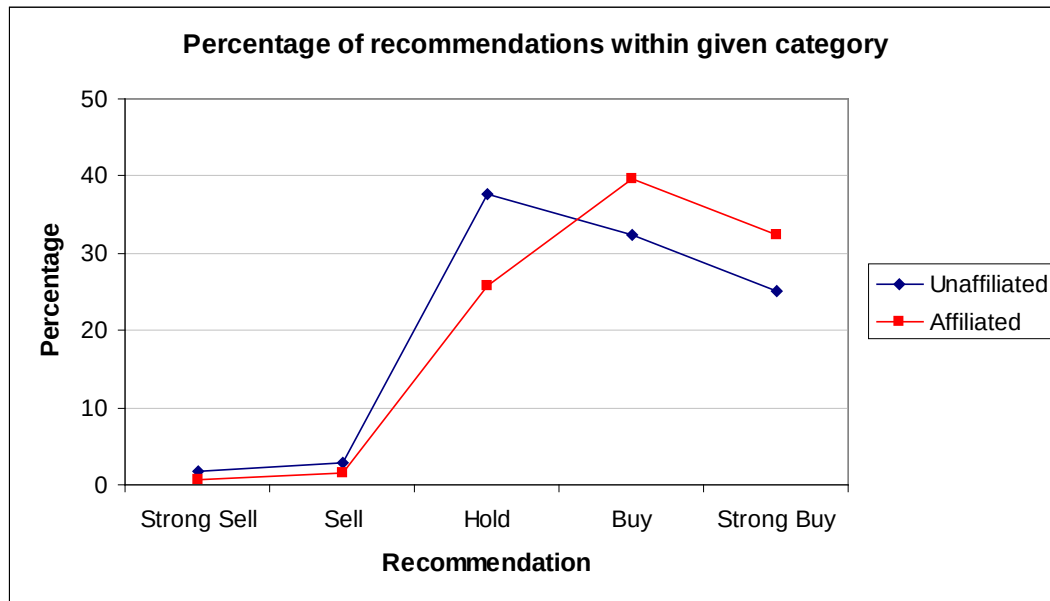
Using the approximations $t_C = t_0$ and $v_C = v_0$, the expression simplifies to $f = ((v_T - v_C) / (e_T - e_C)) * (t_T / (1 - v_C))$. This expression takes into account that the media can affect voting along two margins—by inducing voters to switch party, or by inducing non-voters to turn-out. For DellaVigna and Kaplan (2007), the estimated $v_T - v_C$ is from Table IV, column (4). The additional term $t_T / (1 - v_C)$ equals 1.024. The exposure rate $e_T - e_C$ is from Table VIII, Column (2) using the diary audience measure. For Enikolopov, Petrova, and Zhuravskaya (2007), we estimate v_T and v_C based on Table 1 and Table 3 adding the vote share for the parties OVR, SPS, and Yabloko, and the exposure rate difference $e_T - e_C$ is from Table 7. The additional term $t_T / (1 - v_C)$ equals $.79 / .832 = .949$. For Gerber, Karlan, and Bergan (2009), the Democratic vote share (based on self-reports) in the control group v_C is $.41 / .73 = .56$ (Table 1c). The treatment effect $v_T - v_C$ is .112 (Table 4, Column (4)). The additional term $t_T / (1 - v_C)$ equals $.72 / .44 = 1.64$ (Table 1c) The difference in exposure rate is from page 9 in paper.

For Gentzkow (2006), we take the estimate of t_C from the mean of base-year turnout (Table 1) and combine it with a treatment effect of $10 * -.196 = -1.96$ percent (column (4) in Table 4). The exposure rate difference is an approximation of the share of population with TV exposure from Figure 4. For Gentzkow and Shapiro (2009), we take t_C from page 14 and combine it with the treatment effect of the introduction of one newspaper (Table 2, Column (1)); the exposure rate difference is the estimated effect on the share of population reading a local paper when present (page 14).

Persuading Donors. In all four studies in this Section, a control group with no solicitation is not included in the study. However, we know that unsolicited donations (or survey completion) are very rare, so we approximate them to zero, hence $t_C = 0$ (and $e_C = 0$). For List and Lucking-Reiley (2001), we calculate t_T from Table 1 averaging across the conditions with and without refunds. The exposure rate difference in this study is assumed to be 1 for lack of better measure of how many people read the mailing. For Landry et al. (2006), we aggregate across all treatments (Table 2) and computing the share giving (t_T) and the share

opening the door (e_T). For DellaVigna, List, and Malmendier (2009), we take the share completing the survey (t_T) and the share opening the door (e_T) from Table 4, Treatment NoFlyer-\$0-10 min. For Falk (2009), we take the estimate of t_T from the results in Table 1, and the exposure rate difference is assumed to be 1.

Figure 1. Distribution of stock recommendations by analysts



Notes: The Figure plots the share of recommendations that are are “Strong Sell”, “Sell”, “Hold”, “Buy”, and “Strong Buy” for both unaffiliated and affiliated analysts. Affiliated analysts are analysts that work for an investment bank that has recently undertaken a banking transaction for the company that the analyst covers. Source of data is Malmendier and Shanthikumar, 2009.

TABLE 1
PERSUASION RATES: SUMMARY OF STUDIES

Paper	Treatment	Control	Context	Variable: Persuasion Rate f (Share of Listeners Convinced by Media)								
				Variable t	Year	Recipients	Time Horizon	Sample size	Treatment group t_T	Control group t_C	Exposure rate $e_T - e_C$	Persuasion rate f
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
<u>Persuading Consumers</u>												
Simester et al. (2007) (NE)	17 clothing catalogs sent	12 catalogs	Women's Clothing	Share Purchasing ≥ 1 item	2002	Infrequent Cust. Frequent Cust.	1 year	N = 10,000 N = 10,000	36.7% 69.1%	33.9% 66.8%	100%* 100%*	4.3% 7.0%
Bertrand et al. (forthcoming) (FE)	Mailer with female photo Mailer with 4.5% interest rate	Mailer no photo Mailer 6.5% i.r.	Loan offer in South Africa	Applied for loan	2003	Previous clients of bank	1 month	N=53,194	9.1% 9.1%	8.5% 8.5%	100%* 100%*	0.6% 0.6%
<u>Persuading Voters</u>												
Gosnell (1926)	Card reminding of registration	No card	Presid. Elect.	Registration	1924	Chicago voters	Few days	N = 5,700	42.0%	33.0%	100.0%	13.4%
Gerber and Green (2000) (FE)	Door-to-Door GOTV Canvassing GOTV Mailing of 1-3 Cards	No GOTV No GOTV	Congress. Elect.	Turnout	1998	New Haven voters	Few days	N = 29,380	47.2% 42.8%	44.8% 42.2%	27.9% 100%*	15.6% 1.0%
Green, Gerber, and Nickerson (2003) (FE)	Door-to-Door Canvassing	No GOTV	Local Elect.	Turnout	2001	Voters in 6 cities	Few days	N = 18,933	31.0%	28.6%	29.3%	11.8%
Green and Gerber (2001) (FE)	Phone Calls By Youth Vote Phone Calls 18-30 Year-Olds	No GOTV No GOTV	General Elect. General Elect.	Turnout Turnout	2000 2000	Voters in 4 cities Voters in 2 cities	Few days	N = 4,377 N = 4,278	71.1% 41.6%	66.0% 40.5%	73.7% 41.4%	20.5% 4.5%
DellaVigna and Kaplan (2007)	Availab. of Fox News Via Cable	No F.N. via cable	Presid. Elect.	Rep. Vote Share	'96-'00"	28 US States	0-4 years	N = 66m	56.4%	56.0%	3.7%	11.6% ⁺
Enikolopov, Petrova, and Zhuravskaya (2007) (NE)	Availability of independent anti-Putin TV station (NTV)	No NTV	General Elect.	Vote Share of anti-Putin parties	1999	Russian voters	3 months	N = 45m	19.0%	16.8%	19.9%	10.2% ⁺
Gerber, Karlan, and Bergan (2009) (FE)	Free 10-week subscription to Washington Post	No Subscr.	Governor Elect.	Dem. Vote Share (stated in survey)	2005	Virginia	2 months	N = 1,011	67.2%	56.0%	92.0%	20.0% ⁺
Gentzkow (2006) (NE)	Exposure to Television	No Television	Presid. Elect.	Turnout	40s-'50s	All US Counties	10 years	N = 100m*	54.5%	56.5%	80.0%	4.3%
Gentzkow and Shapiro (2009)	Exposure to Local Newspaper	No local paper	Presid. Elect.	Turnout	1868-1940	All US Counties	0-4 years	N = 100m*	67.8%	67.0%	46.0%	5.1%
<u>Persuading Donors</u>												
List and Lucking Reiley (2001) (FE)	Fund-raiser mailer with low seed Fund-raiser mailer with high seed	No mailer No mailer	Campaign for computer lab	Share Giving Money	1999	Florida donors	1-3 weeks	N = 1,000 N = 1,000	3.7% 8.2%	0% 0%	100%* 100%*	3.7% 8.2%
Landry et al. (2006) (FE)	Door-To-Door Fund-raising Campaign for University Center	No visit	Campaign	Share Giving Money	2004	North Carolina Donors	immediate	N = 4,833	10.8%	0%	36.3%	29.7%
DellaVigna et al. (2009) (FE)	Door-To-Door Survey	No visit	Surveys	Share Completing	2008	Chicago	immediate	N = 607	10.3%	0%	41.4%	24.8%
Falk (2009) (FE)	Fund-raiser mailer with no gift Mailer with gift (4 post-cards)	No mailer No mailer	No gift	Share Giving Money	2004	Swiss donors	1-3 weeks	N = 3,237 N = 3,347	12.2% 20.6%	0% 0%	100%* 100%*	12.2% 20.6%

Notes: Calculations of persuasion rates by the authors. The list of papers indicates whether the study is a natural experiment ("NE") or a field experiment ("FE"). Columns (9) and (10) report the value of the behavior studied (Column (4)) for the Treatment and Control group. Column (11) reports the Exposure Rate, that is, the difference between the Treatment and the Control group in the share of people exposed to the Treatment. Column (12) computes the estimated persuasion rate f as $100 \cdot (t_T - t_C) / ((e_T - e_C) \cdot (1 - t_C))$. The persuasion rate denotes the share of the audience that was not previously convinced and that is convinced by the message. The studies where the exposure rate (Column (11)) is denoted by "100%*" are cases in which the data on the differential exposure rate between treatment and control is not available. In these cases, we assume $e_T - e_C = 100\%$, which implies that the persuasion rate is a lower bound for the actual persuasion rate. In the studies on "Persuading Donors", even in cases in which an explicit control group with no mailer or no visit was not run, we assume that such a control would have yielded $t_C = 0\%$, since these behaviors are very rare in absence of a fund-raiser. For studies with vote share as dependent variable (denoted with a "+"), see Appendix A for the expression for the persuasion rate. For additional details, see Appendix A